LAST NAME: _____

Practice Quiz 11A — MATH 2055 — Fall 2026

Practice for Quiz 11

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Prepared using GPT-6 in the Codex harness.

1. Let $C = \{n^3 : n \in \mathbf{Z}^+\}$. Define a function $f : \mathbf{Z}^+ \to C$ and prove that it is a bijection. What does this say about the cardinalities of C and $\mathbf{Z}^+$?